

tu 17

Vincent Wang

September 2026

1 Differential Forms

Claim: Pullback of sum and product. Let $F : N \rightarrow M$ be C^∞ , $g \in C^\infty(M)$, and $\omega, \tau \in \Omega_1(M)$. Then $F^*(\omega + \tau) = F^*\omega + F^*\tau$ and $F^*(g\omega) = (F^*g)(F^*\omega)$.

Proof: Let $p \in N$ and $X_p \in T_pN$. Then

$$\begin{aligned}(F^*(\omega + \tau))_p(X_p) &= (\omega + \tau)_{Fp}(F_*X_p) = \omega_{Fp}(F_*X_p) + \tau_{Fp}(F_*X_p) \\ &= F^*(\omega)(X_p) + F^*(\tau)(X_p) \\ (F^*(g\omega))_p(X_p) &= (g\omega)_{Fp}(F_*X_p) = g(Fp)\omega_{Fp}(F_*X_p) = (F^*g)(p)(F^*\omega)_p(X_p) \\ &= ((F^*g)(F^*\omega))_p(X_p)\end{aligned}$$

Prop 17.12. The pullback $F^*\omega$ of a C^∞ 1-form ω on M by a C^∞ map $F : N \rightarrow M$ is a C^∞ 1-form on N .

Sps $f : \mathbb{R} \rightarrow S^1 \leq \mathbb{R}^2$ is given by $f(t) = (\cos t, \sin t)$. If $\omega = -ydx + xdy$ on S^1 , compute the pullback $f^*\omega$.

Ans: $(f^*\omega)(dt) = \omega_{fp}(f_*(dt|_p)) = \omega_{fp}(-\sin t dx + \cos t dy) = 1$, so $f^*\omega = dt$.